

1) Fuel Kernel: This is the fissile material where fission takes place in the particle

Buffer: Porous layer the mechanically separates the fuel kernel from the IPyC layer and accommodates fission products and kernel swelling.

Inner Pyrolytic Carbon: This layer protects the SiC pressure boundary layer from fission product attack and, with the OPyC, holds the SiC in a compressive state

SiC: The SiC layer is the pressure vessel and fission product barrier

Outer Pyrolytic Carbon: This layer protects the SiC during manufacturing and, with IPyC, holds the SiC in compression.

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2) Oxide fuel kernels are easier to fabricate and have better fission product retention than carbide fuel kernels. when fission occurs, oxygen is released which forms oxides with fission products. However, these oxide-forming fission products deplete and then oxygen potential increases leading to CO production and kernel migration. Carbide fuel kernels do not produce CO like oxide fuel kernels, but are poor at retaining fission products which can lead to damage of the buffer, carbon, and SiC layers. To balance these trade-offs, an 80/20 oxide/carbide mixture can be used to lower oxygen potential while also retaining fission products.

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3) There are 5 groups of fission products present in TRISO particles, and each acts differently:

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a) Soluble oxides: Usually rare earth metal oxides, these dissolve in fuel matrix

b) Insoluble oxides: Form insoluble oxides in the fuel lattice

c) Metals: Pd and Ag are of concern. These can precipitate and interact with layers outside the fuel kernel.

d) Noble Gases: These are insoluble and form voids

e) Volatiles: Can be solid or condensed depending and often include halogens and transition metals.

- part of it ...

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4) Below the critical amorphization temperature black spot defects predominate and strain from these defects leads to amorphization. Swelling in this regime increases logarithmically until the entire structure amorphizes.

Above the critical amorphization temp and below 1273K, there is a point defect regime. Here, swelling increases logarithmically until saturation where the rates of defect formation and destruction are the same. As temperature increases, the saturation swelling decreases because defects annihilate more readily at higher temperatures. Above 1273K, Frank loop formation dominates in a void swelling regime. next
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In this regime, vacancies come together to form voids and there is no saturation swelling level.

Higher dose and higher temperature both lead to increased swelling in this regime.

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5) Irradiation leads to degradation of the thermal conductivity within the SiC layer. In the point defect regime, phonons are scattered by defects which degrades conductivity. The effect saturates. Degradation in the void swelling regime does not saturate, but is less severe than point defect degradation. The primary scatterers are grain boundaries and point defects.

- needed a bit more

b) Graphite undergoes a phenomenon called turn around under irradiation.

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When nuclear graphite is manufactured, its built-in anisotropy leads to small microcracks called Mrzowski cracks. This is due to weaker van der Waals bonding in the "c" direction and much stronger covalent bonding in the "a" direction. Irradiation displaces carbon atoms in the graphite leading to contraction in the "a" direction. In response, the overall structure decreases in size as the "c" direction swells and fills the already-present Mrzowski cracks. Eventually, all the cracks become filled and the overall structure begins to swell. This change from shrink to swell is called turn around.

It is important to recognize that irradiation creep is fundamental to relaxing the stresses that form during dimensional shifts in nuclear graphite.

- wanted specific interstitial/vacancy effects

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7) a) Overpressure occurs when gases build up in the buffer leading to excessive stress on the SiC layer and ultimate fuel failure. Buffer fabrication issues and too high of burnup can cause this.

b) Fission product attack occurs when elements like Pd or Ag migrate out of the fuel kernel and buffer. They react with the SiC layer leading to thinning and failure. CO attack on the PyL layers is another example of fission product attack

c) $\text{IP}_{\text{yC}}/\text{SiC}$ ~~debonding~~ occurs when the IP_{yC} layer shrinks under irradiation. As the layers separate, the SiC layer enters tension and is no longer strong which ~~leads~~ to cracking and particle failure.

8) One example of advanced TRISO is using ZrC instead of ~~SiC~~.

5/5 ZrC has much ~~better~~ high temperature performance and fission product protection (no reaction with Pd). It is worth mentioning that ZrC has worse ~~oxidation~~ performance than SiC.

9) IHX materials are susceptible to corrosion from impurities in He such as CO_2 , H_2O , and CH_4 .

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Alloy 617 is the common IHX material and contains a chromium oxide layer which aids in corrosion resistance when there is too much carbon activity, carburization can occur where excess carbides form leading to embrittlement. On the other end of the spectrum, at low carbon activity, decarburization occurs which leads to poor creep strength.

- almost all